
Understanding representation learning in RL-based logic optimization

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Abstract

1 Computer chips ought to be as small as possible for cost efficiency. The problem
2 of minimizing chip size while preserving the chip’s functionality has been studied
3 through the prism of and-inverter graphs (AIGs) since any boolean function can be
4 represented by AIGs. Because the complexity of AIG size minimization is unknown
5 (it is NP-hard under some assumptions), it is a fitting candidate for neural solvers
6 that can leverage learned representations to generalize to unseen AIGs and outpace
7 standard enumerations and branch-and-bound solvers. Reinforcement learning and
8 more generally planning have been used to optimize such AIGs without the need
9 for intractable supervision signals. Contemporarily, graph representation learning
10 have been applied to train models that can perform tasks on unseen AIGs. More
11 recently, those two technologies have been combined and achieved competitive
12 performances with state-of-the-art heuristics. In this preliminary work, we question
13 the adequacy of graph representation learning to improve deep reinforcement
14 learning for logic AIG minimization.

15 1 Introduction

16 **Contributions:** First, we show that when replicating past work’s graph reinforcement learning
17 pipelines, graph representation learning did not help the training compared to simple multi-layer
18 perceptrons trained with graphs statistics (number of nodes, edges, ...). More fundamentally,
19 we show that both theoretically and empirically many graph representations, even AIG-specific
20 ones, cannot predict the values of AIGs—in the RL sense—more accurately than standard multi-layer
21 perceptrons trained on graphs statistics only.

22 2 Related work

23 2.1 Combinational optimization

24 2.2 Graph representation learning for AIGs

25 (GraphBench, DeepGate, ...)

26 2.3 Reinforcement learning for AIGs

27 (AlphaSynth, MLCAD, ...)

28 2.4 Graph reinforcement learning for other combinatorial optimization

29 (review paper ...)

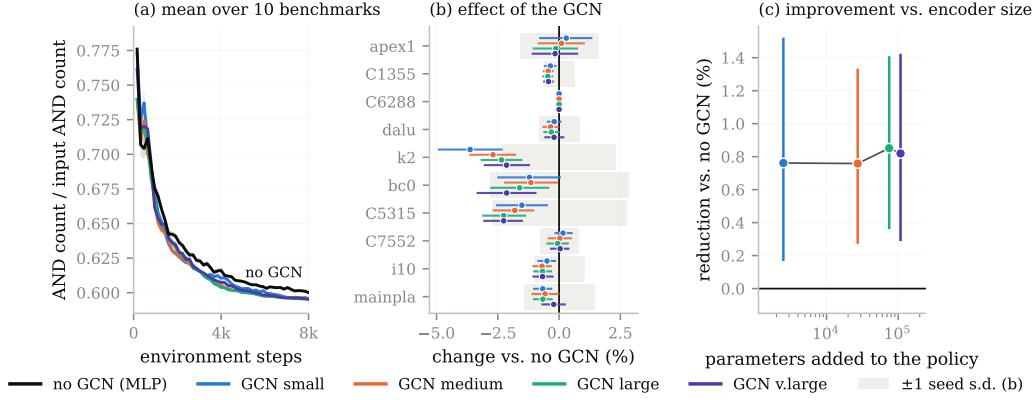


Figure 1: PPO on the ten MLCAD benchmarks with five observation encoders, hyper-parameter matched. (a) mean AND count over the ten benchmarks; (b) per-benchmark change against the MLP, shaded band is one seed standard deviation; (c) mean reduction against the parameters the encoder adds to the policy.

3 Background material

3.1 And-inverter graphs minimization

3.2 Markov decision process

4 Methodology

4.1 Graph state representation

4.1.1 Graph convolutional networks

4.1.2 Engineered graph features

4.2 Reinforcement learning

5 Experiments

5.1 Graph reinforcement learning

Adding a GCN to the policy does not improve AIG minimization (Figures 1, 2). We train PPO on ten MLCAD benchmarks with five observation encoders — an MLP on graph statistics and four GCNs spanning 2,496 to 107,968 added parameters — over a grid of three learning rates, three entropy coefficients and five seeds, for 2,250 runs in total. Encoders are compared only on the 424 of 450 (benchmark, learning rate, entropy, seed) cells that all five completed, so no encoder is tuned against the others. Every GCN reduces the final AND count by 0.76–0.85% relative to the MLP, which is less than the 1.23% standard deviation induced by changing the seed alone, and a $43\times$ increase in encoder size moves that figure by nothing.

The result holds over a hundred further circuits and does not grow with circuit size (Figures 3, 4). We repeat the comparison on ex200–ex299, a suite spanning 106 to 46,977 input AND gates. The three GCNs reduce the final AND count by 0.33–0.71% against a reseed standard deviation of 0.54%, and the effect is flat across two and a half orders of magnitude of circuit size. We note that the MLP runs were executed on CPU and did not finish on the largest circuits, which is why 20 benchmarks are excluded.

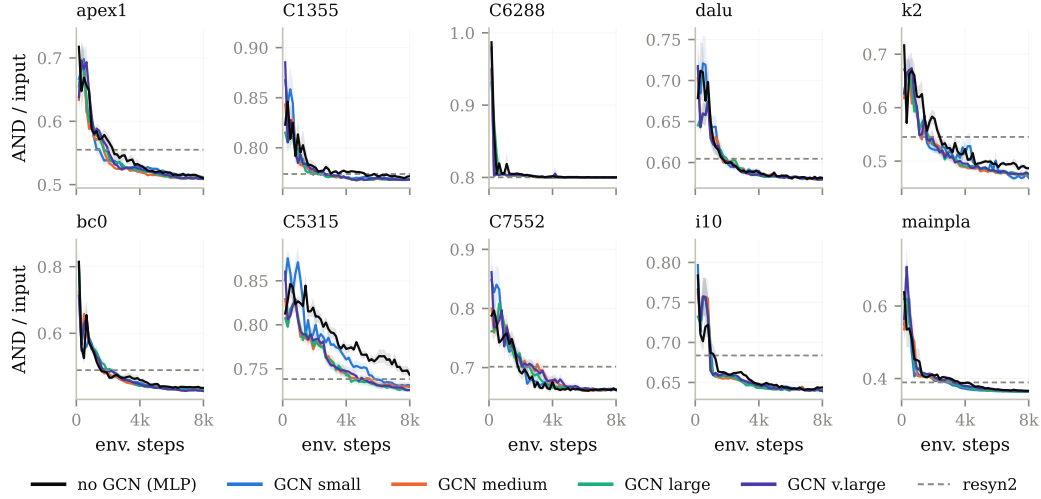


Figure 2: Per-benchmark learning curves for the runs aggregated in Figure 1(a). Dashed line: `resyn2`.

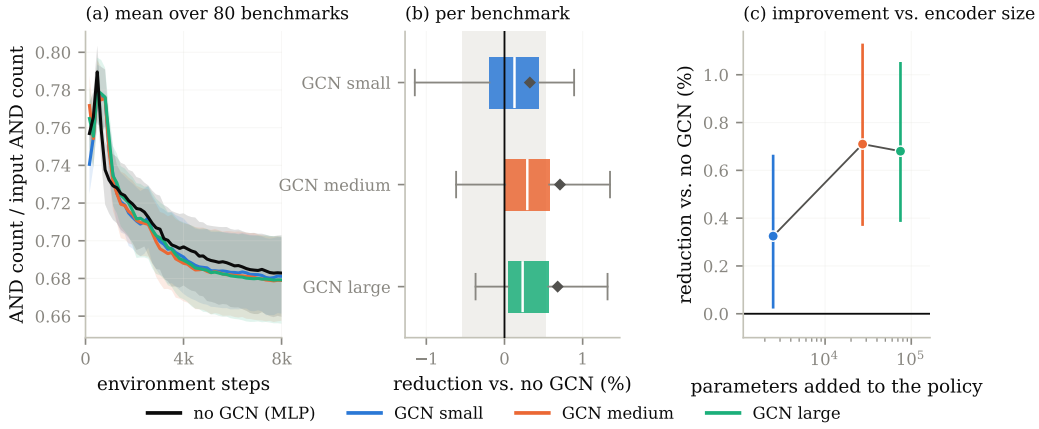


Figure 3: The same comparison on `ex200–ex299`, where each encoder ran at its own baseline hyper-parameters. Restricted to the 390 of 500 (benchmark, seed) cells all four encoders completed, covering 80 benchmarks.

5.2 Value prediction

5.2.1 Building a dataset

5.2.2 Theoretical lower bounds on the prediction error

Message passing cannot predict V^* better than the statistics the MLP already sees (Figures 5, 6). A k -layer MPNN separates graphs at most as finely as k iterations of 1-WL, so grouping states by their 1-WL colour and predicting each group’s mean V^* gives an error no MPNN can beat; the same construction bounds a 0-layer network, a (size, depth) model, and a model seeing only the MLP’s features x_G . Message passing is worth a great deal over size and depth alone, but the MLP’s own features close almost all of that gap: the two bounds are within a factor of 1.7 on eight of nine benchmarks and within 1.1 on five, and rank states equally well ($\tau \approx 0.98$). Depth is irrelevant — the bound moves by at most 1.1×10^{-3} between $k = 1$ and $k = 50$, and edge direction and inverter bits change it by less than 10^{-6} . C6288 is excluded: its target takes two distinct values.

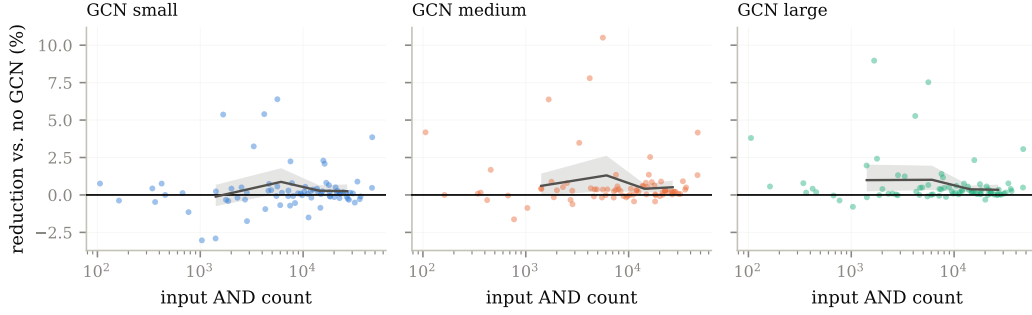


Figure 4: Reduction against the MLP as a function of input circuit size on ex200–ex299. Line: mean over size quartiles.

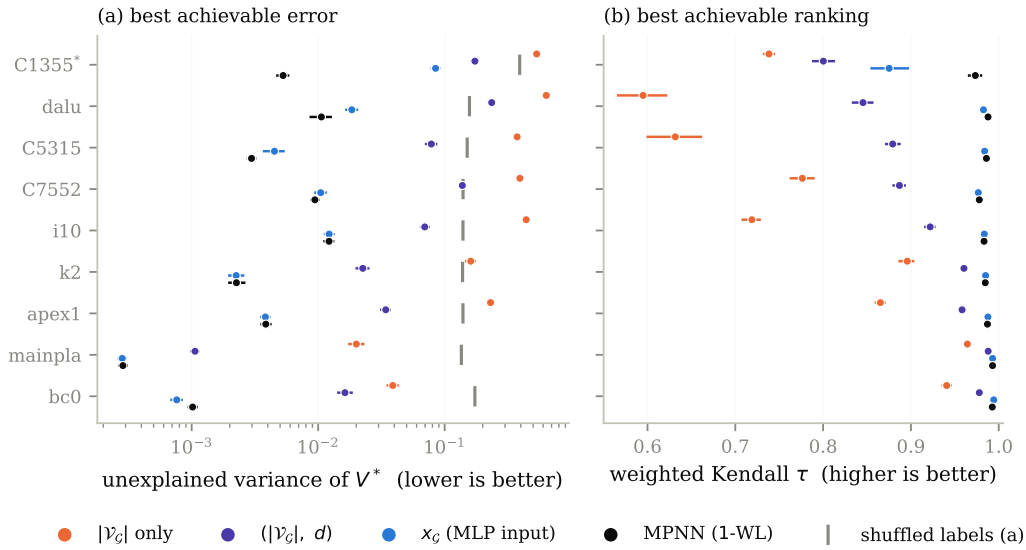


Figure 5: Best error (a) and best ranking (b) achievable by each model class on V^* , over nine benchmarks and ten dataset seeds. Each class is replaced by the optimal predictor for the information it can distinguish, so the value shown cannot be beaten by any model in that class. *C1355 has only six distinct targets.

5.2.3 Empirical performances of AIG-specific models on value prediction

Trained graph models, including AIG-specific ones, do not beat predicting the mean (Figures 7, 8). We train four generic message-passing networks (GCN, GIN, GAT, GraphSAGE), three AIG-specific ones (DeepGate, PolarGate, FuncGNN) and an MLP on x_G , each over its own grid of 36 configurations (12 for DeepGate), selecting on validation error and reporting test error. All seven graph models land between 0.90 and 1.74 times the target variance — at or above the constant predictor — while the MLP reaches 0.47 at the median. Their training error is also ≈ 1 , so this is a failure to fit rather than to generalise, and it leaves them two to three orders of magnitude above the bound of Figure 5 that their own architecture class admits.

6 Future work

Many actions

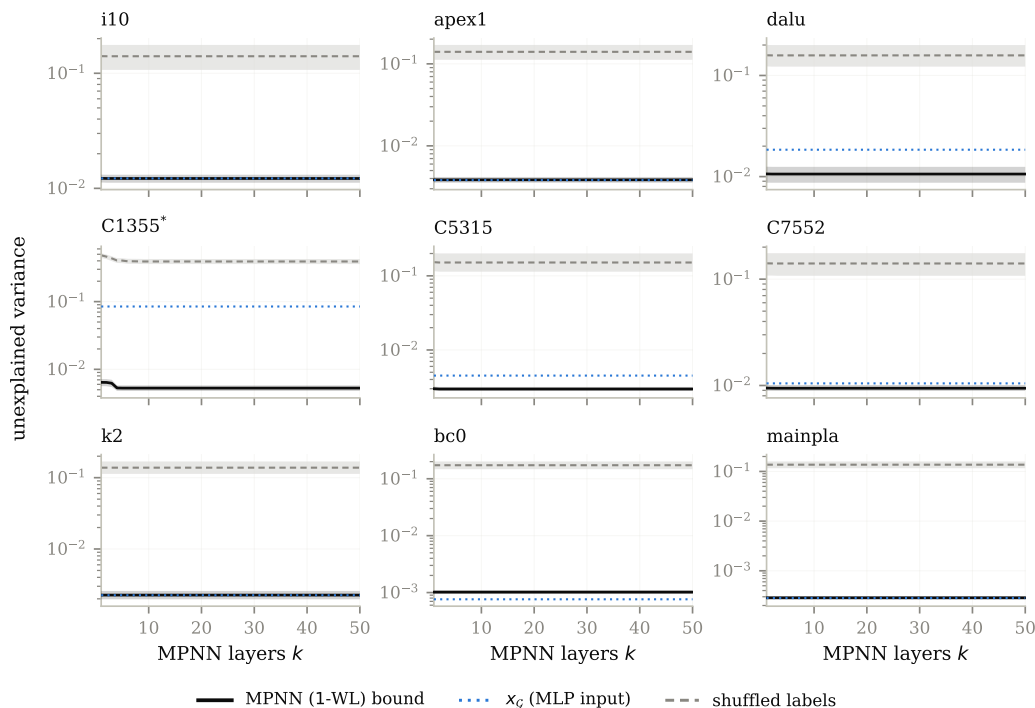


Figure 6: The MPNN bound as a function of depth k .

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- [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural Simulation System*. New York: TELOS/Springer-Verlag.
- [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

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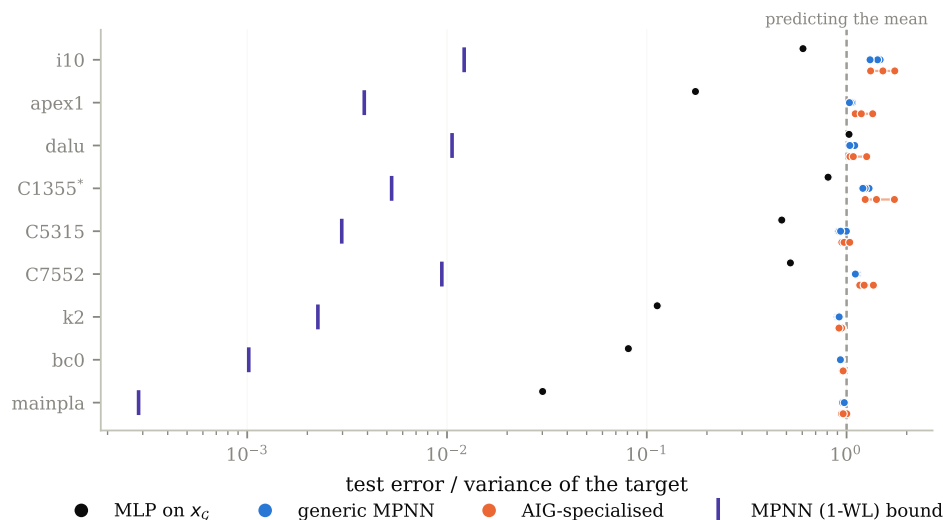


Figure 7: Test error of eight trained models, normalised by the variance of the target so that 1.0 is the error of predicting the mean. Violet ticks: the 1-WL bound of Figure 5.

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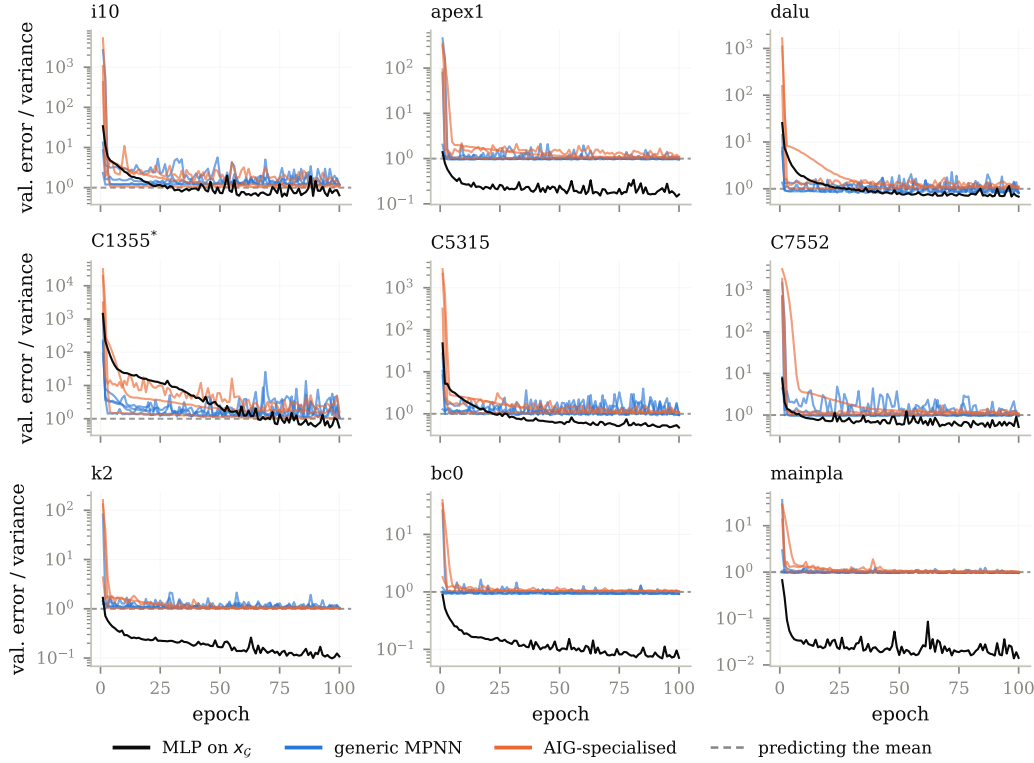


Figure 8: Validation error over training for the configurations of Figure 7.

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